

Seam Carving: A Review and Experiment

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Abstract

Seam carving, a content-aware image resizing technique, has gained significant attention in computer graphics and image processing domains. It can be a good solution to the problem of distortion and distortion caused by inconsistent changes in the aspect ratio of width and height when resizing an image. This paper provides a comprehensive review of seam carving, presenting related work and algorithms, discussing its technical challenges, summarizing key findings, and shows some experimental results. The study aims to enhance the understanding of seam carving's principles, technical implementation, and practical applications.

1. Introduction

This part will explain some backgrounds and concepts related to seam carving, and will illustrate the basic ideas of the seam carving algorithm.

1.1. Background

Image resizing is widely applied across various fields. In web design, image resizing is used to adapt to different screen sizes and resolutions, ensuring a consistent user experience. In mobile applications, image resizing is employed to fit various mobile device screens. Image editing software utilizes scaling functions for image adjustments and modifications to meet specific requirements. In printing and publishing, image resizing is used to accommodate different layout designs and printing needs. In computer vision and machine learning, image resizing is used to standardize input image sizes, enhancing algorithm performance. Multimedia and video streaming applications employ image resizing to adapt to different display screens. Digital signage and display systems utilize image scaling to ensure accurate content presentation on large screens. In medical imaging, image resizing is used to adjust the size of medical scan images for accurate diagnosis and analysis.

Therefore, the image resizing algorithm has become an important technique in the field of image processing. Traditional image scaling methods typically employ uniform scaling or simple interpolation algorithms, which may result in distortion and loss of image content. Content-aware image resizing aims to achieve better resizing results by preserving the important content structure of an image while reducing scaling on less significant information.

Traditional resizing methods do not differentiate between important and less important regions in an image, which can lead to deformation or loss of crucial content. The simplest resizing algorithm, for example, scales proportionally according to the aspect ratio of the original image, with the effect shown in Fig. 1b. Furthermore, how about resizing by cropping? As shown in Fig. 1c, this looks worse because not all images have the right size crop option. Content-aware image resizing, on the other hand, analyzes the content features of an image, such as edges, textures, and color distributions, to determine which regions are important. Based on their significance, the resizing is applied non-uniformly, minimizing the impact on important areas while allowing more resizing or removal of less important regions.

1.2. Seam Carving Algorithm

The key challenge of content-aware image resizing lies in accurately identifying the important content in an image and performing effective resizing. To address this efficiently, Avidan and Shamir [1] proposed a content-aware image resizing algorithm, called seam carving, which can change the size of an image by elegantly carving-out or inserting pixels in different parts of the image. Seam carving calculates the energy value of each pixel in an image, finds the seam path with the lowest energy (a path connecting the top and bottom of the image), and removes or expands it to achieve image resizing. The effect of the seam carving algorithm is shown in Fig. 1d.

The basic idea of the algorithm is intuitive. First we calculate the energy map of the image. The energy of each pixel is usually calculated based on factors like gradients,

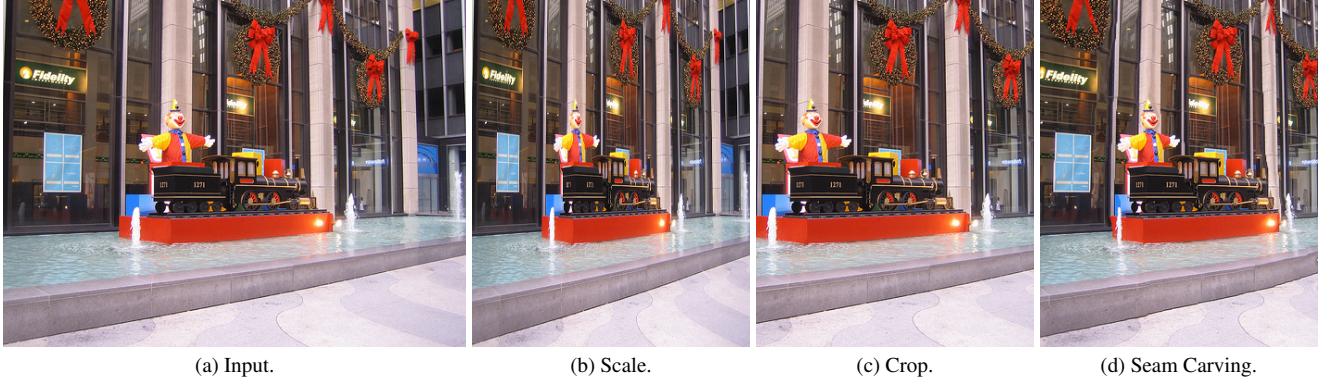


Figure 1. Comparison of the results of different scaling methods.(Images from Avidan and Shamir’s [1] original paper)

color differences or texture variations. The energy map represents the significance of each pixel in the image. Next, It uses dynamic programming techniques, considering the energy values of neighboring pixels to determine the seam path with the lowest cumulative energy, i.e. the optimal path. Once the lowest-energy seam is identified, it is either removed or duplicated. Seam removal involves eliminating the pixels along the path, while seam duplication replicates the pixels. This process is iteratively applied to resize the image according to the desired dimensions. For downsizing, multiple low-energy seams are removed, and for upsizing, seams are duplicated or inserted. Finally, the remaining pixels are resized to maintain the continuity and visual quality of the image. Interpolation or resizing techniques can be used to fill in gaps created by seam removal, or duplicated to fit an enlarged area. The detailed algorithms are described in Part 3.

2. Related Work

Over the last few decades, many algorithms have been proposed in relation to content-aware image resizing. In the area of content-aware cropping, Suh et al. [14] proposed a method that computes an importance map and utilizes a greedy approach to search for a rectangular cropping window. The window is selected to encompass the maximum percentage of salient points in the image. Ma et al. [10] introduced a method that involves segmenting the image using fuzzy k-means clustering based on color and texture. Saliency is then computed considering the relative location and area of the regions, as well as the entropy of the source image. Zhang et al. [18] presented a method for auto-cropping based on an energy function. The energy function incorporates three terms: composition, conservative, and penalty. To find an optimal solution, the objective function is maximized using particle swarm optimization (PSO) algorithm. Santella et al. [13] proposed a method that utilizes gaze points for image cropping. The image is

divided into regions based on color similarity, and a saliency map is generated using gaze tracking data. The fixation data is employed to identify salient content and determine the optimal cropping results. To achieve optimal cropping, a cost function is minimized to include the region of interest, maximize important areas in the image, and avoid cutting through background objects.

In the area of content-aware scaling, Jiang et al. [6] proposed an efficient edge-adaptive scaling algorithm that segments the input image into different types of image blocks using a directional edge detector. Interpolation is applied along the direction of the edges. Liang et al. [9] presented a patch-wise scaling method that focuses on the salient area of the image while preserving the global visual effect. Local bidirectional similarity and smoothness measures are used to evaluate the quality of the retargeted image. The source image is segmented into important and unimportant patches based on the saliency map.

In the other areas of content-aware resizing. Yu-Shuen Wang et al. [16] present a scale-and-stretch warping method for content-aware image resizing that preserves visually prominent features. Their approach computes optimal local scaling factors iteratively and utilizes a significance map to guide the amount of deformation, minimizing the impact on important regions. The method distributes distortion in all directions, enabling efficient interactive resizing. Guo-Xin Zhang et al. [17] propose a novel image resizing method that aims to preserve important local regions and image edge structure. they define handles to describe these regions and edges, assigning weights based on an importance map. By minimizing a weighted sum of quadratic distortion energies, their method diffuses distortion in all directions and preserves important edges. It is efficient and offers a closed-form solution.

Based on seam carving, several studies have improved and extended the original algorithm. Rubinstein et al. [12] focus on video retargeting using an improved seam carving operator. They remove 2D seam manifolds from 3D

space-time volumes using graph cuts, and introduce a novel energy criterion to enhance visual quality. The approach considers the energy introduced by the operator and applies forward-looking seam removal. Weiming Dong et al. [4] present a method that combines seam carving and image scaling for content-aware image resizing. The principle behind their method is the use of a bidirectional similarity function of image Euclidean distance (IMED), while cooperating with a dominant color descriptor (DCD) similarity and seam energy variation. In applying the seam carving algorithm to stereo images, Kei Utsugi et al. [15] present a novel technique for content-preserving stereo image editing by extending seam carving. Their approach addresses the challenge of maintaining consistency between the left and right images during the seam carving process, enabling virtual manipulation of object depths in the scene while preserving the overall content. Apart from this, Tali Dekel Basha et al. [2] introduce a method for retargeting a pair of stereo images while preserving the geometric structure and 3D perception of the scene. By extending single image seam carving to stereo images, the proposed method minimizes visual and depth distortion. It considers visibility relations between pixels, ensuring geometric consistency with the original scene.

Daniel Domingues et al. [3] present Stream Carving, a novel content-aware image resizing scheme based on seam carving. Their approach introduces larger seams ("streams") and recovers resulting holes using inpainting, resulting in improved quality compared to the original method and comparable performance to other image retargeting techniques. There is, of course, work dedicated to improving the speed of algorithm execution. Jinseok Lee et al. [8] propose two methods to accelerate the original seam carving algorithm for image resizing while preserving important information. One method involves partial updates to the cumulative minimum energy map (CMEM) around the seam, while the other method uses a divide and conquer (DNC) approach to process multiple seams simultaneously. These methods improve the speed of the algorithm by 2 to 5 times while maintaining the effectiveness of preserving image information. Stephan Kopf et al. [7] present the FSCAV algorithm (Fast Seam Carving for Size Adaptation of Videos) to resize videos for small mobile device displays. The algorithm extends seam carving from images to videos, preserving essential content without visible artifacts. It offers low computational complexity and addresses distortions by applying alternative adaptation techniques for shots with low visual quality. User evaluations validate the approach's high visual quality. Ronald Duarte and Resit Sendag [5] introduce an accelerated approach for content-aware resizing of images and videos using the seam carving algorithm. By leveraging CUDA-enabled GPUs, the implementation achieves significant speed-ups compared to

CPU-based methods, enabling real-time resizing of high-resolution videos and large images. To address the computational time of seam carving for content-aware image retargeting, Dipti Patel et al. [11] propose an accelerated approach that removes or inserts multiple pixel-wide batch seams in a single iteration. The width of the batch seam is adaptively adjusted to preserve image energy, while minimizing inserted energy.

3. Algorithm

According to the paper, the seam carving algorithm is executed in the following steps:

1. Energy calculation:

First, for the input image, the energy value of each pixel needs to be computed. The energy value reflects the importance or information content of the pixel and guides the selection of pixels to be removed. Typically, the energy value is calculated based on pixel gradients (gradient energy) or other features such as contrast. The gradient energy can be obtained by computing the brightness variation of each pixel in the horizontal and vertical directions. For example, the gradients of each pixel can be calculated using Sobel or Scharr operators, and the horizontal and vertical gradients can be weighted and summed to obtain the energy value.

2. Energy map construction:

Next, the computed energy values are used to construct an energy map. The energy map is a 2D array with the same width and height as the original image, where each element represents the energy value of the corresponding pixel.

3. Dynamic programming for computing the optimal energy path:

Using dynamic programming, the cumulative energy for each pixel is calculated starting from the bottom of the energy map. For each pixel, its cumulative energy is compared with the cumulative energies of adjacent pixels in the previous row, and the path with the minimum energy is selected. This process is repeated row by row, resulting in the determination of the optimal energy path from the bottom to the top.

4. Removal or copy of pixels along the optimal energy path:

Finally, based on the computed optimal energy path, the corresponding pixel rows or columns are removed or copied, depending on the current task is enlargement or reduction.

The previous operations 1-4 are repeated until the desired image size is achieved. The flow diagram is shown in Fig. 2.

The key of the seam carving algorithm lies in the dynamic programming process, which determines the pixels to be removed by calculating the optimal energy path. Due to the computational complexity of energy calculation and dynamic programming, optimization techniques are often employed to speed up the algorithm, such as using fast dynamic programming, approximate dynamic programming, or parallel computing methods. This is something we will discuss later.

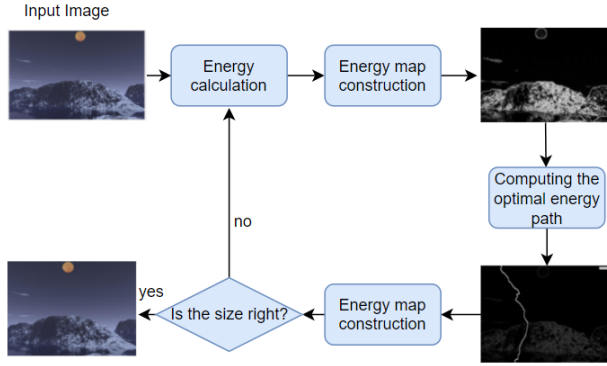


Figure 2. Seam carving algorithm execution flow.

3.1. Energy calculation and Energy map construction

Assuming that the input image I is of size $m \times n$. First calculate the energy map of the image, the energy map is generally the gradient modulus of the image pixels, in order to simplify the calculation can first convert the image to grayscale image. Usually The energy function is defined as the gradient in the x direction plus the gradient in the y direction (there are of course other ways of defining the energy function, which I will not go into here.):

$$e_1(I) = \left| \frac{\partial}{\partial x} I \right| + \left| \frac{\partial}{\partial y} I \right| \quad (1)$$

A vertical seam is defined as:

$$\begin{aligned} s^x &= \{s_i^x\}_{i=1}^n = \{(x(i), i)\}_{i=1}^n \\ s.t. \forall i, |x(i) - x(i-1)| &\leq 1 \end{aligned} \quad (2)$$

where x is a mapping $x : [1, \dots, n] \rightarrow [1, \dots, m]$. That is, a vertical seam is an 8-connected path of pixels in the image from top to bottom, containing only one pixel in each row of the image. Similarly, if y is a mapping $y : [1, \dots, m] \rightarrow [1, \dots, n]$, then a horizontal seam is:

$$\begin{aligned} s^y &= \{s_j^y\}_{j=1}^m = \{(j, y(i))\}_{j=1}^m \\ s.t. \forall j, |j(i) - j(i-1)| &\leq 1 \end{aligned} \quad (3)$$

Given an energy function e (i.e. Eq. (1)), we can define the cost of a seam as:

$$E(s) = E(I_s) = \sum_{i=1}^n e(I(s_i)) \quad (4)$$

3.2. Dynamic programming for computing the optimal energy path

We want to look for the optimal seam s^* that minimizes this seam cost (Eq. (4)) :

$$s^* = \min_s E(s) = E(I_s) = \min_s \sum_{i=1}^n e(I(s_i)) \quad (5)$$

The optimal seam can be determined through dynamic programming. The initial step involves traversing the image from the second row to the last row, computing the cumulative minimum energy, denoted as M , for all possible connected seams at each entry (i, j) :

$$\begin{aligned} M(i, j) &= e(i, j) + \\ &\min(M(i-1, j-1), M(i-1, j), M(i-1, j+1)) \end{aligned} \quad (6)$$

Upon completion of this procedure, the minimum value in the last row of M will signify the termination point of the minimal connected vertical seam. The computation of M for horizontal seams follows a similar definition.

3.3. Technical difficulties and optimizations

Considering that the authors did not present the performance, technical difficulties and optimisation methods of the algorithm in the original paper, I focus on this part in this section.

Here are the main technical challenges of the Seam carving algorithm:

- Energy function design:

Seam carving algorithm requires an energy function to determine the pixel energy in the image. The energy function measures the differences between pixels and their surrounding pixels to aid in determining the optimal carving path. Designing an accurate and effective energy function is a key technical challenge because it needs to consider various features and structures in the image to avoid introducing distortions in important regions.

- Carving path search:

Finding the optimal carving path is the core task of the Seam carving algorithm. This involves searching for a path with the minimum energy throughout the entire image. Typically, dynamic programming or

graph theory algorithms are used to solve this problem. Dynamic programming methods are often employed to compute the minimum energy path at each pixel, while graph theory methods represent the image as a graph and find the optimal path using shortest path algorithms.

- Pixel deletion and insertion:

Seam carving algorithm adjusts the size of the image by removing or inserting pixels. When deleting pixels, an entire row or column of pixels needs to be completely removed from the image, while inserting pixels requires the insertion of new rows or columns of pixels. This involves pixel shifting and reallocation to maintain image continuity and smoothness. Efficient handling of these operations is necessary in order to avoid introducing unnecessary artifacts or distortions.

- Algorithm efficiency:

Seam carving algorithm involves energy calculations and path searches throughout the entire image, which can be computationally intensive. For large images or cases that require multiple adjustments, algorithm efficiency becomes an important consideration. Optimization techniques such as memoization or Floyd algorithm in dynamic programming or image pyramids can be employed to improve the efficiency of the algorithm.

There are several common methods for optimizing the Seam Carving algorithm:

- Dynamic programming: Traditional Seam Carving algorithms use dynamic programming to calculate the optimal seam (minimum energy path), but it can be computationally expensive for large images. Optimization methods include approximate dynamic programming or spatial optimization to reduce computational complexity.
- Parallel computing: Utilizing parallel computing techniques such as GPUs (Graphics Processing Units) or multi-core CPUs can speed up the execution of Seam Carving algorithms. Performing simultaneous calculations for multiple seams or utilizing parallel computing for energy calculations can significantly improve algorithm performance.
- Fast energy calculation: Traditional Seam Carving algorithms compute energy for each pixel, which can be a bottleneck. Optimization techniques include fast energy calculation methods, such as leveraging local properties of the image, utilizing integral images, or using approximate energy calculation methods to reduce computational overhead.

- Priority queues: Seam Carving algorithms typically use priority queues to select the best seam. Managing and updating the priority queue efficiently using efficient data structures and algorithms can reduce the algorithm's time complexity.

- Edge preservation: In the process of image resizing, it may be necessary to preserve important edges or objects. Optimization methods include integrating edge information into energy calculations or using edge preservation constraints to adjust seam selection.

3.4. Applications

Now let's summarise the applications of the seam carving algorithm. As mentioned above, the seam carving algorithm can be used to reduce image height and width or to increase image height and width. The effect of resizing is shown in Fig. 3.

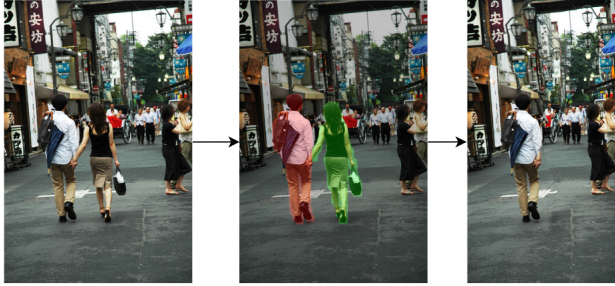


Figure 3. Application: resizing.

In addition, the seam carving algorithm can be used to remove certain areas from an image or to protect areas from being removed. The seam carving algorithm can be used for object removal by modifying the energy matrix of the image. By setting the energy of specific pixels to a very low level (i.e. using a mask), the seam carving method can eliminate the targeted object (Fig. 4a). This is achieved by reweighting the energy matrix in the region where the object appears. The user selects the object to be removed, and seams are removed from the image until all marked pixels are eliminated. The system automatically determines the



(a) The original image is shown on the left and the image after the removal of the girl is shown on the right.



(b) Protection of the man in the image and removal of the effect of the woman.

Figure 4. Application: remove and protect.

appropriate vertical or horizontal removals based on the target removal region. Similarly, using a mask to set the energy of specific pixels to a very high level will ensure that the pixels in this region will not be removed or deformed due to the insertion of the seam (Fig. 4b).

4. Experiment

To test the performance of the seam carving algorithm, I implemented the seam carving algorithm using python and experimented with image shrinking and image enlarging.

4.1. Image Shrinking

The image resizing algorithms mentioned in this article are also commonly used in social media to allow images to be displayed on different devices, and images on social media can be broadly classified into three types: landscapes, photos of objects and group photos of people. I have therefore tested all three types of images.

4.1.1 Landscape

First let's see how the seam carving algorithm performs on landscape photos. As shown in Fig. 5a, this is a relatively simple landscape of a small fishing boat sailing down a river. Obviously the boat is the focus of the image and more of it should be retained when performing image shrinking. The results of the algorithm execution are shown in Fig. 5b and Fig. 5c, and as expected the seam carving algorithm performs well, retaining the most important elements of the image.

Next I tested this with a more complex landscape. As shown in Fig. 6a, this is a photograph of a snowy mountain



(a) The original image.



(b) Seam carving 1.



(c) Seam carving 2.

Figure 5. Image Shrinking: fishing



(a) The original image.



(b) Seam carving.

Figure 6. Image Shrinking: mountain

and a lake, and it is clear that the focus of the image should be the view in the middle. seam carving algorithm was executed as shown in Fig. 6b, and it is clear that the important elements of the image were retained.



(a) Origin.



(b) Seam carving.

Figure 7. Image Shrinking: object.

4.1.2 Object

When an image contains an object, that object is usually the focus of the image and should not be overly affected when shrinking the image. As shown in Fig. 7a, it is clear that the bird is the focus of the image and the seam carving algorithm is executed as shown in Fig. 7b, achieving the desired result.

4.1.3 Group Photos of People

The application of the seam carving algorithm to the group photo shown in Figure 1 encountered some obstacles, as the gradient information in the image is more complex due to the density and complexity of the people, making it difficult for the algorithm to determine what is "unimportant information". The results of the algorithm are shown in Figure 3, where the leftmost figure is very much reduced (perhaps because it is an obnoxious character?).

4.2. Image Enlarging

As mentioned previously, the seam carving algorithm has the capability to both reduce and expand the size of an image. When expanding an image, we compute the average value of the optimal seam along with its neighboring pixels. This average value is then duplicated along the op-



(a) Origin.



(b) Seam carving.

Figure 8. Image Shrinking: Group Photos of People.



(a) Origin.



(b) Seam carving.

Figure 9. Image enlarging: evening.

timal seam to add a new row or column. To avoid repeatedly copying the same seam, wherein each specific seam is considered optimal, we can first identify the top K optimal seams in the original image. These K seams can then be copied simultaneously, resulting in an increase of K rows or columns. I made image enlarging on Figure 1 and obtained the result shown in Figure 2, which shows good results and a very natural image.

5. Conclusion

In this article, I extensively researched and discussed about seam carving algorithm. In Sec. 1, I introduced the

concept, background and applications of the seam carving algorithm. In Sec. 2, I present some scientific work related to image resizing and to the seam carving algorithm, including extended applications of the algorithm. In Sec. 3, I explain in detail the whole process of the seam carving algorithm. In addition I present the technical difficulties, optimisation directions and application scenarios of the algorithm. In Sec. 4, I tested the seam carving algorithm's ability to handle different scenes under different conditions. As a classical computer vision algorithm, its performance is still quite good.

I have learnt a lot through this work and have become more impressed with the seam carving algorithm. In addition, my research skills have been honed in the process of completing my thesis.

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